

BT302 - FINAL TERM SUBJECTIVE MEGA FILE

ALL DATA FROM PAST PAPERS

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1. Affinity maturation

The affinity of antibodies depends upon the dosage of antigen, the low dosage of antigen causes increase in the affinity of antibodies which is called as affinity maturation.

A secondary response can elicit antibodies with several fold greater affinity than in a primary response. Affinity maturation primarily occurs on membrane immunoglobulin of germinal center B cells and as a direct result of somatic hyper mutation (SHM) and selection by T cells.

2. Antibodies types

There are 5 types of heavy chain constant regions in antibodies (immunoglobulin) and according to these types; they are classified into IgG, IgM, IgA, IgD, and IgE. They are distributed and function differently in the body.

3. AIDS (10)

Acquired immunodeficiency syndrome (AIDS) is a chronic, potentially life-threatening condition caused by the human immunodeficiency virus (HIV). By damaging your immune system, HIV interferes with your body's ability to fight infection and disease.

HIV is a sexually transmitted infection (STI). It can also be spread by contact with infected blood and from illicit injection drug use or sharing needles. It can also be spread from mother to child during pregnancy, childbirth or breastfeeding. Without medication, it may take years before HIV weakens your immune system to the point that you have AIDS.

Symptoms

The symptoms of HIV and AIDS vary, depending on the phase of infection.

Causes

HIV is caused by a virus. It can spread through sexual contact, illicit injection drug use or sharing needles, contact with infected blood, or from mother to child during pregnancy, childbirth or breastfeeding.

4. Alternate pathway (10)

The alternative pathway is a type of cascade reaction of the complement system and is a component of the innate immune system, a natural defense against infections. The alternative pathway is one of three complement pathways that opsonize and kill pathogens. The pathway is triggered when the C3b protein directly binds a microbe. It can also be triggered by foreign materials and damaged tissues.

Functions of Complement Proteins

Complement system provides specific as well as non-specific resistance against infections. This kind of resistance is exhibited with following two important functions

Primary function

This function of complement proteins is considered primary as these activated proteins used to kill or lyse infectious agents like bacteria. For this complement proteins get deposited on the surface of infectious agents which make hole in those cells for their lysis or killing.

Secondary or Effector functions

These functions of complement proteins effect on other functions of immune system.

Complement Activation

Complement proteins are activated through a series of cascading reactions which are called as complement activation. Followings are the important pathways for complement activation

- 1) Classical Pathway
- 2) Alternate Pathway
- 3) Mannose Binding Lectin (MBL) Pathway

These pathways generate the activated form of complement like C5 which leads towards a pathway called as common pathway as a result the membrane attack complex is generated for the lysis of microorganism.

5. Cellular immunity

Cellular mediated immunity provides protection by following ways:

T-cell immunity

In this form of protection, cytotoxic T-cells (CTL) are activated for the killing of infected and transformed cells.

By activating macrophages

By this way, macrophages are activated for destroying various pathogens.

By activating natural killer (NK) cells

In this way of protection, NK cells are activated for killing of transformed and viral infected cells.

6. Name of barriers against infection pathogens?

Innate immune system is the first line of defense against invading pathogens. It is comprised of following three important components or barriers

- 1) Anatomical barriers
- 2) Humoral (Secretory) barriers
- 3) Cellular barriers

7. Somatic Hyper mutation (2)

Somatic Hyper mutation

The phenomenon of enhanced rate of point mutation in the immunoglobulin V region genes is called as Somatic hyper mutation. It occurs particularly in V gene which codes for 2nd hyper variable region. As a result, there is an increase in immunoglobulin diversity after various antigenic stimulation. Moreover, it also causes increase the affinity of immunoglobulin in order to compete for limited amount of available antigens.

8. Application of agglutination reaction (3)

Application of agglutination reactions

Agglutination reactions are used for following ways

- Blood group determination: ABO & Rh for both either on the basis of antigens of RBC or antibodies of serum
- Assessing bacterial infections e.g Widal test for diagnosis of Typhoid fever

9. Define onco-fetal antigen and it's types (3)

The tumor antigens which are also expressed by normal cells. These are expressed in high levels by tumor cells as compare to normal cells. These kinds of antigens are also called as **onco-fetal antigens** as these are expressed during early development & lost during adult life. However, these antigens re-expressed on tumor cells. These antigens also serve as tumor markers due to their importance in the diagnosis & prognosis of cancers

There are following two types of onco-fetal antigens

- 1) Alpha-fetoproteins (AFP)
- 2) Carcino-embryonic antigens (CEA)

10. Hypersensitivity and it's types (10)

It is the state of undesirable reaction by normal immune system, in which immune system overreacts against various stimuli. These are also known as hypersensitivity reaction or intolerance. In hypersensitivity, the protective role of immune system becomes harmful for its host. This condition is also termed as Allergy which is an abnormal response against the otherwise harmless environment stimulus e.g food, pollen & animal dander. The hypersensitive state of immune response also exhibits in the form of autoimmune disorders which are the abnormal immune response against host's self- tissues.

Classification of Hypersensitivity

Hypersensitivity reactions are classified on the basis of following two factors:

- 1) Nature of immune response involved
- 2) The time required for immune response

Followings are the important types of hypersensitivity reactions

i) Type I

These reactions are also known as Immediate or Anaphylactic reactions which involve various tissues of the body with following clinical conditions

- Skin: Urticaria & Eczema
- Eyes: Conjunctivitis
- Nose: Rhinorrhea, Rhinitis
- Lungs: Asthma
- Gastro-Intestinal Tract (GIT): Gastroenteritis

These reactions manifest various undesirable symptoms from minor inconveniences to fetal death. In these reactions, usually reaction is quick & immediate ranging 15 to 30 minutes, however, sometimes can be 10-12 hours.

ii) Type II

This type of hypersensitivity is also known as cytotoxic hypersensitivity. This condition affects various organs & tissues of body which determines the tissue specific nature of hypersensitivity. For such reactions there is the involvement of various antigens. These antigens are usually either endogenous (from inside of body) like antigens of body's self-tissues or exogenous (from outside of body) like various chemicals in the form of haptens.

The example of endogenous antigens induced cytotoxic hypersensitivity include erythroblastosis fetalis which is also called as hemolytic disease of newborns (HDN). This condition occurs due to Rh incompatibility between mother & fetus. As a result, there is the development of anti-Rh antibodies in maternal serum (IgG) which has the ability of crossing placenta in subsequent pregnancy.

iii) Type III

These reactions are also known as immune complex mediated hypersensitivity. The reaction time required for such reactions is 3-10 hours after exposure to antigen. There is generation of antibodies which bind with corresponding antigens as a result soluble immune complex formation occurs. These immune complexes affect generally the host and that condition is called as serum sickness. Otherwise, these reactions are specific to various organs and cause the following conditions

- Skin: Systemic Lupus erythematosus (SLE), Arthus reaction
- Kidney: Lupus nephritis
- Lung: Aspergillosis
- Blood vessels: Polyarteritis
- Joints: Rheumatoid Arthritis

iv) Type IV

These reactions are also termed as cell mediated hypersensitivity. Due to more time requirement like 48-72 hours, these reactions are also called as delayed type of hypersensitivity. Immunologically, T-cells, Macrophages and monocytes are involved in these reactions.

In these reactions, the response is against intracellular pathogens that's why these reactions are antibody independent in nature. Lymphocytes like CD8 (Cytotoxic) T-cells cause direct damage to target cells. Likewise, CD4 (Helper) T-cells for example Th1 cells recognize foreign antigen with MHC class II. Moreover, the secretion of Cytokines, activation of macrophages and inflammation increase the magnitude of tissue injury & damage.

11. Pathway of complement activation (10)

Classical Pathway of Complement Activation

In this pathway, the complement proteins become activated by antibodies attached on the surface of pathogen (bacteria). C1 protein is activated after interacting with Fc region of antibodies IgG or IgM. After activation of C1 complex, C2 & C4 are activated. Finally C3 proteins are activated after the cleaving action of C2 & C4 in a cascade manner

Alternate Pathway of Complement Activation

In this pathway, there is no need of antibody for the activation of complement proteins. Rather C3 is directly converted into activated form. However, various co-factors of serum facilitate the

activation of C3 sequentially. After fully activation of C3 proteins, finally C3 helps in the lysis of bacterial cell

Lectin Pathway of Complement Activation

Lectin pathway is considered very similar to classical pathway in context to activation of complement protein 1 (C1). In place of antibodies, mannoses on the surface of specific pathogens are involved in activation of C1. Additionally, Mannose binding Lectins (MBL) in serum bind to surface of pathogen containing mannose in their cell wall e.g Fungus. Finally, C3 proteins are produced after interaction of mannose & Lectins which further make membrane attack complex for lysis of infectious agent.

12. Ways of extracellular killing?

Extracellular pathogens are dispatched by a combination of direct killing (NK cells and complement) and ingestion (macrophages and neutrophils).

13. Immune cells of myeloid lineage

The myeloid progenitor is the precursor of the granulocytes, macrophages, dendritic cells, and mast cells of the immune system. Macrophages are one of the three types of phagocyte in the immune system and are distributed widely in the body tissues, where they play a critical part in innate immunity.

14. Diagnose of type 1 hypersensitivity

Diagnosis of Type I Hypersensitivity

Type I hypersensitivity reactions are diagnosed in the following important ways

- **By Skin (Prick & Intradermal) test**

In this test, the allergen is administered through skin and the reaction is observed in the form of redness, itching and swelling of the skin

- **Measurement of total IgE levels**

In patient's serum, the measurement of total IgE also help in the diagnosis of type I hypersensitivity reactions.

- **Allergen specific IgE levels**

In addition to total IgE levels, the allergen specific IgE levels also help in the diagnosis of immediate hypersensitivity reactions

15. How complement proteins remove extracellular pathogens from blood?

Complement proteins perform several functions. They serve as a marker to indicate the presence of a pathogen to phagocytic cells, such as macrophages and B cells, to enhance engulfment. This

process is called opsonization. Certain complement proteins can combine to form attack complexes that open pores in microbial cell membranes. These structures destroy pathogens by causing their contents to leak. When innate mechanisms are insufficient to clear an infection, the adaptive immune response is informed and mobilized.

16. Natural and Artificial immunization?

Natural active immunization

Natural immunization is acquired immunization which is achieved by natural exposure to antigen or acquiring infection by an infectious agent or microorganism e.g measles virus. This is also termed as natural active immunization. This form of natural exposure to antigen stimulate the humoral immunity as a result specific antibodies are generated against exposed microbe. Moreover, there is also activation of cell mediated immunity for intracellular pathogen.

Artificial Active Immunization

Artificial immunization is the kind of immunization which is acquired by artificial exposure to antigen or microorganism which is administered to body in less virulent or attenuated form like heat killed forms of microbes. This artificial exposure with antigen stimulate the humoral immunity in the form antibodies generation. Moreover, artificial activation of cell mediated immunity for intracellular pathogen for example BCG for Mycobacterium tuberculosis. Artificial immunization: acquired immunization by artificial exposure. This artificial exposure to antigen or microbe which is by administration of less virulent (attenuated), heat killed microbes to organism. Artificial exposure to antigen stimulate the humoral immunity for antibodies

Natural Passive Immunization

On the other hand, natural immunization can be acquired immunization which can be obtained by isolating gamma globulins (antibodies) from other's individual. This is called as natural passive immunization. Furthermore, the transfer of gamma globulins (IgG) from mother's blood to fetus via placenta. The placental cells contains FcR γ for IgG transfer to fetal circulation. Similarly, the natural transfer of IgA can be occur through breast milk to new-born baby.

Artificial Passive immunization

This form of artificial immunization is the acquired immunization which is achieved by administering gamma globulins or antibodies from others individual's or immune animal's serum for example antibodies against tetanus

17. Signal for t-cell activation?

For the activation of T-cells following two signals are required

1. Signal 1 is generated after engagement of TCR with Antigen/MHC complex
2. Signal 2 is generated after engagement of supporting molecules with their corresponding ligands

18. How pre-cytotoxic cell activate t-lymphocytes?

The surface of a cytotoxic T-cell has something called a CD8 receptor. The CD8 receptor interacts with cells called major histocompatibility complex (MHC) Class I molecules to recognize when healthy cells are infected. When the CD8 receptor recognizes an infected cell, it activates the cytotoxic T-cells.

19. Uses of artificial passive immunization and complications associated with it?

This form immunization is done by transferring or injecting of specific gamma-globulins from other individual or from other immune animals. Artificial passive immunity is used against various following acute bacterial infections

- ▪ Diphtheria
- ▪ Tetanus

Moreover, artificial passive immunity is also used against various poisoning conditions like

- Insects or sting biting e.g immunization against various strains of Malaria producing Plasmodium
- Reptiles e.g anti-venom administration in case of snake biting
- Food poisoning (Botulism)

Artificial passive immunity is also used as prophylactic (preventive) measure against various viral infections like Influenza, Poliomyelitis, Measles and Rabies.

20. Lectin pathway?(5 or 10) Marks

Lectin pathway is considered very similar to classical pathway in context to activation of complement protein 1 (C1). In place of antibodies, mannose on the surface of specific pathogens is involved in activation of C1. Additionally, Mannose binding Lectins (MBL) in serum bind to surface of pathogen containing mannose in their cell wall e.g Fungus. Finally, C3 proteins are produced after interaction of mannose & Lectins which further make membrane attack complex for lysis of infectious agent.

Effector Functions of Complement Proteins

Complement proteins have following effector functions for providing immunity against infectious agents

- **Opsonization (Opsonins)**

Various complement proteins bind with the surface of pathogens which facilitate the process of phagocytosis by phagocytes. This enhanced process of phagocytosis is called as opsonization, while complement proteins involved in this process are called as opsonins.

- **Chemoattraction (Chemoattractants)**

Complement proteins also served as chemoattractants for the movement of phagocytic cells towards the site of infection. The chemoattractant function of complement proteins increased the defense mechanism against invading agents.

- **Anaphylaxis (Anaphylatoxins)**

Complement proteins also involved in hypersensitivity (allergic) reactions which are also called as anaphylactic reactions in the form of exaggerated immune response. Such kind of complement proteins are called as anaphylatoxins.

- **Pro-Inflammatory**

The role of complement proteins in inflammatory process is well established. The tissue response against infectious agent is termed as inflammatory response and the complement proteins contributing towards such kind of response are considered as pro-inflammatory.

21. Factors influencing immunogenicity, it's contribution, contribution of biological methods and method of administration? 10Marks

FACTORS INFLUENCING IMMUNOGENICITY

Contribution of the Immunogen

Foreignness

The immune system normally discriminates between self and non-self such that only foreign molecules are immunogenic.

Size

There is not absolute size above which a substance will be immunogenic. However, in general, the larger the molecule the more immunogenic it is likely to be.

Chemical Composition

Generally, the more complex the substance is chemically the more immunogenic it will be. The antigenic determinants are created by the primary sequence of residues in the polymer and/or by the secondary, tertiary or quaternary structure of the molecule.

Physical form

In general particulate antigens are more immunogenic than soluble ones and denatured antigens more immunogenic than the native form.

Degradability

Antigens that are easily phagocytosed are generally more immunogenic. This is because for most

antigens (T-dependant antigens) the development of an immune response requires that the antigen be phagocytosed, processed and presented to helper T cells by an antigen presenting cell (APC).

Contribution of the Biological System

Genetic Factors

Some substances are immunogenic in one species but not in another. Similarly, some substances are immunogenic in one individual but not in others (i.e. responders and non-responders). The species or individuals may deficient or have altered genes that code for the receptors for antigen on B cells and T cells or they may not have the appropriate genes needed for the APC to present antigen to the helper T cells.

Age

Age is another biological factor which can also influence immunogenicity. Frequently the very young and the very old have a reduced ability to mount an immune response in response to an immunogen.

Method of Administration

Dose

The dose of administration of an immunogen can influence its immunogenicity. Generally the optimal dose of immunogen considered as having good immunogenicity. There is a dose of antigen above or below which the immune response will not be optimal.

Route

Generally the subcutaneous route which is underneath the skin is better than the intravenous or intragastric routes. The route of antigen administration can also alter the nature of the response

Adjuvants

Substances that can increase the immune response to an immunogen are called adjuvants. The use of adjuvants, however, is often hampered by undesirable side effects such as fever and inflammation.

22. Define Hapten carrier conjugate?

Hapten-carrier conjugate is an immunogenic molecule to which hapten is non-immunogenic while carrier is immunogenic in nature. The binding between hapten and carrier molecule is covalent in nature. Structurally, this conjugate is composed of haptenic & carrier determinants. The carrier determinants are native in nature as these have to generate immune response

23. Write characteristics of immunoglobulin antibodies

Immunoglobulins are also called as antibodies. These are the component of secretory or humoral part of specific immunity. These are glycoprotein molecules that are produced by plasma cells in

response to an immunogen and which function as antibodies. The immunoglobulins derive their name from the finding that they migrate with globular proteins when antibody-containing serum is placed in an electrical field. Immunoglobulins have following important characteristics or functions

Primary Functions

The binding of specific antigen with its corresponding antibody is the primary function of immunoglobulin. This function determines the specific nature of antibodies. Moreover, the binding sites on antibodies for specific antigenic determinants are called as paratopes. Each antibody has its own specific paratope for its specific antigen. The number of available antigen binding sites on antibody is called as valency of antibody. Each antibody has minimum valency of two as two antigenic determinants can bind with a single antibody molecule.

Secondary or Effector Functions

After binding with its specific antigen, the antibody performs its secondary functions which are also called as biological or effector functions. These are as follows

- 1) Complement activation: complement proteins bind with antibody after binding with its corresponding antigen which leads towards complement activation for their effector function.
- 2) Binding to various cell receptors via specific receptors to Fc fragment of antibody also called as Fc receptors (FcR). These receptors present on the surface of phagocytic cells like neutrophils & macrophages for performing their corresponding effector functions

24. Define somatic recombination.

Recombination involves rearrangement of DNA in somatic cells. The newly recombined genes are not inherited in contrast to germ cells. Primary Ig repertoire differs slightly from one individual to next one. Also differ in individual's lifetime by their exposure to different antigens. As B-cell differentiates into mature one, it makes light chain (Kappa). There is rearrangement of genes (exons) as introns are removed and genes start to express. In this recombination, V genes become close to J genes by removing introns. Recombined DNA processed into mature RNA by splicing. As Bcell differentiates into mature one, it make heavy chains. There is rearrangement of genes (exons) as introns are removed and genes start to express. In this recombination, D genes become close to J & then V recombines with DJ. Finally recombined DNA processed into mature RNA by splicing

25. Difference in endogenous Ab and exogenous Ab antibodies?

Antigen Processing & Presentation of Endogenous antigens

The process in which the antigen is being processed is called as antigen processing. This process varies according to the source of antigen. The intracellular source of antigen which is also called as Endogenous antigen reside in the cytosol of antigen presenting cell like Viruses. For

presenting such kind of endogenous antigens MHC I is required which is present in antigen presenting cells. MHC I express on all nucleated cells of body.

Antigen Processing & Presentation of Exogenous antigens

Unlike the endogenous antigens, exogenous antigens are not cytosolic in nature while reside in the endosome e.g bacteria which are taken up by the process of endocytosis in endosomes. Class II MHC is required for presenting such kind of antigens. MHC II express on limited number of cells like antigen presenting cells (APC) like macrophages, dendritic cells & B-cells. In contrast to endogenous antigens, exogenous proteins are processed in endosomes by proteolytic proteins like proteases.

26. Usage of monoclonal antibodies

Monoclonal Antibodies

The antibodies which are produced from a single antibody producing B-cell are called as monoclonal antibodies. These antibodies also able to bind with single and unique antigen binding site (epitope). Monoclonal antibodies mainly consist of single subtype of IgG for example IgG1, IgG2 & IgG3.

27. Write different steps involved in phagocytosis.

Phagocytosis can be divided into four main steps:

- (i) recognition of the target particle
- (ii) signaling to activate the internalization machinery
- (iii) phagosome formation, and
- (iv) phagolysosome maturation.

28. Immune cells of adaptive immunity (2)

Adaptive immune responses are carried out by white blood cells called lymphocytes. There are two broad classes of such responses—antibody responses and cell-mediated immune responses, and they are carried out by different classes of lymphocytes, called B cells and T cells, respectively.

29. Primary and secondary immune response (2)

The first contact that an organism has with a particular antigen will result in the production of effector T and B cells which are activated cells that defend against the pathogen. The production of these effector cells as a result of the first-time exposure is called a **primary immune response**.

The **secondary immune response** is one of the most important features of immune systems. During the secondary immune response, the immune system can eliminate the antigen, which has been encountered by the individual during the primary invasion, more rapidly and efficiently.

30. Characteristics of Antibody (5)

Antibody formation process is well controlled mechanisms with following significant characteristics

- 1) Self/Non-Self Discrimination
- 2) Memory
- 3) Specificity

31. What is MHC. Differences btw MHC I and MHC II (5)

In adaptive immunity, cell to cell interaction or cell mediated immunity is one of the important mechanism for providing defense against pathogens. In cell mediated immunity, cell to cell interaction is being orchestrated by a structure called as immunological synapses. This synapse is generated during interaction between antigen presenting cells (APC) and T lymphocytes. From Tcells the T-cell receptor (TCR) contributes for this synapse. While from antigen processing cells (APC), MHC with processed peptides contribute for this synapse. There are following two classes of major histocompatibility complex (MHC)

- 1) Class I MHC
- 2) Class II MHC

The MHC expressed on all nucleated cells of body. T-cells are grouped functionally on the basis of associated

MHC with fragmented protein fragments

- 1) Cytotoxic T-Cells which recognize antigens in association with class I MHC
- 2) Helper T-Cells which recognize antigens in association with class II MHC

32. What is the structure of class II MHC molecules 2

Class II MHC is composed of two polypeptide chains α & β chains of equal length. These chains are anchored in the plasma membrane through Trans membrane. Both chains also have a cytoplasmic region for phosphorylation & binding to cytoskeleton of cell.

The α & β chains contain a highly conserved $\alpha 2$ & $\beta 2$ domain for binding to one of the accessory molecule of T-cell known as CD4. Similarly, a highly polymorphic peptide binding region formed by $\alpha 1$ & $\beta 1$ domains where antigen binds. This antigen binding groove accommodates antigen peptides.

33. Write the names of pathway complement activation system 2

- 1) Classical Pathway
- 2) Alternate Pathway

34. Which factors affect the antibody and antigen test 3

The good antigen & antibody reactions are known due to high specific & sensitive nature. Various factors are involved which determine the nature of antigen and antibody reactions which are as follows

- 1) Affinity
- 2) Avidity
- 3) Antigen & Antibody Ratio
- 4) Physical form

35. What should be chemical nature of good immunogen 3

Proteins

The majority of immunogens are proteins. These may be pure proteins or they may be glycoproteins or lipoproteins. In general, proteins are usually very good immunogens .

Polysaccharides

Pure polysaccharides and lipopolysaccharides are good immunogens.

36. Write a note on Hybridoma formation 5

The fusion of lymphoid tumor cell with normal B-Lymphocyte with single specificity hybrid cell line is called as hybrid cell line. This hybrid cell line has dual function.

- 1) Immortality of tumor cell
- 2) The production of antibody with single specificity

37. diversity of T cells

The diverse array of TCR is based on the gene which encodes for TCR. Structurally, the TCR gene is composed of various gene segments which are joined together. T-cell maturation occurs in an organ which is called as thymus where the expression of the gene for TCR occur. The germ line gene for α chain contain only two segments which are called as V (Variable) & J (Joining) segments. However, the germ line gene for $\alpha\beta$ chain is composed of three segments called as V (variable), D (diversity) & J (Joining) segments.

The diversity for specificity of TCR is based on combinational diversity which in fact is based on for various combinations of V, D & J segments for TCR. In majority, the specificity of TCR is based on the combination of α & β chains which constitutes the major T-cell population. However, the small population T-cells also exist which carry the TCR with γ (Gamma) & δ (Delta) chains. This subset of T-cells is called as $\gamma\delta$ T-cells.

38. application of monoclonal antibody

The specific monoclonal antibodies can be used against tumor antigen can be served for

- a) As a vehicle for delivering anti-cancer drug to tumor or cancer

b) Activation of components of innate immune system like complement proteins

39. Epitope

Epitope or Antigenic Determinant

That portion of an antigen that combines with the products of a specific immune response

40. characteristics of antibody

Immunoglobulins are also called as antibodies. These are the component of secretory or humoral part of specific immunity. These are glycoprotein molecules that are produced by plasma cells in response to an immunogen and which function as antibodies. The immunoglobulins derive their name from the finding that they migrate with globular proteins when antibody-containing serum is placed in an electrical field

41. Self-antigen and autoimmunity

The specific immunological unresponsiveness or non-reactivity to an antigen resulting from a previous exposure to same antigen is called as **immunological tolerance**. This kind of unresponsiveness is mostly against self-antigens. This form of non-reactivity can be induced for non-self (foreign) antigen. The antigens which induce tolerance are called as **tolerogen**.

Physiologically, there is no immune response against self-antigens which is also termed as self-tolerance. On the other hand, if immune response is generated against self-antigens is called as **autoimmunity (Pathology)**. Immune system recognizes self-antigens and mount strong immune response. The discrimination between self and non-self-antigens is achieved by a process called as **self MHC recognition**. The tolerance can be generated against non-self-antigens

42. characteristic of neutrophils

The neutrophil is the most ubiquitous leukocyte in a normal blood smear, with about one neutrophil for every 1000 red blood cells. They are the smallest in size of the granulocytes. Neutrophils have a characteristic multilobed nucleus, with 3 to 5 lobes joined by slender strands of genetic material. The cytoplasm of neutrophils contains numerous purplish granules called azurophilic or primary granules that contain microbicidal agents. Neutrophils also contain smaller, secondary granules that house lysozyme, gelatinase, collagenase and many other enzymes.

43. usage of monoclonal antibodies

Usage of Monoclonal Antibodies

Monoclonal antibodies have very significant applications in clinical and basic medical sciences which are as follows

1. Usage in prevention, diagnosis & treatment of disease

2. Immunophenotyping of Immune cells, Monoclonal antibodies are used against cell surface molecules like Cluster of Differentiation (CD)
3. Typing of various tumors e.g Leukemia
4. Immunohistochemistry of solid tumors
5. Use in Immunotherapy: Against various tumor marker e.g CD20 B-Cell lymphoma
6. Use in Fluorescence Activated cell sorting (FACS)

44. What is lymphoid organs?

Primary lymphoid organs: These organs include the bone marrow and the thymus. They create special immune system cells called lymphocytes. Secondary lymphoid organs: These organs include the lymph nodes, the spleen, the tonsils and certain tissue in various mucous membrane layers in the body (for instance in the bowel).

45. Mono clonal antibodies (long + short)

Monoclonal Antibodies

The antibodies which are produced from a single antibody producing B-cell are called as monoclonal antibodies. These antibodies also able to bind with single and unique antigen binding site (epitope). Monoclonal antibodies mainly consist of single subtype of IgG for example IgG1, IgG2 & IgG3.

Production of Monoclonal Antibodies

The production of monoclonal antibodies is done using immortal clone of cell with single antibody specificity. The immortal cells are those cells which don't have the ability to proliferate indefinitely. For this purpose, the normal antibody producing cell is being fused with an appropriate B-cell tumor line which is called as hybrid cell. Such kind of hybrid cells are for production of large amount of antibodies.

Usage of Monoclonal Antibodies

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2. Immunophenotyping of Immune cells, Monoclonal antibodies are used against cell surface molecules like Cluster of Differentiation (CD)
3. Typing of various tumors e.g Leukemia
4. Immunohistochemistry of solid tumors
5. Use in Immunotherapy: Against various tumor marker e.g CD20 B-Cell lymphoma

6. Use in Fluorescence Activated cell sorting (FACS)

46. B-cell activation and T-cell antigen 5 marks

B cell activation occurs in the secondary lymphoid organs (SLOs), such as the spleen and lymph nodes. After B cells mature in the bone marrow, they migrate through the blood to SLOs, which receive a constant supply of antigen through circulating lymph.

A T-cell receptor recognizes antigen in the form of a complex of a foreign peptide bound to an MHC molecule. 3-12. T cells with different functions are distinguished by CD4 and CD8 cell-surface proteins and recognize peptides bound to different classes of MHC molecule.

47. Causes of mast cell when releases from cell 3 marks

Mast cells can be activated or triggered by things like:

- Infections.
- Medications.
- Insect venom.
- Reptile venom.
- Fragrances.
- Stress.
- Exercise.
- Food.

48. Oxygen dependent or Oxygen independent intercellular 5 marks

Oxygen independent

In this process of intracellular killing of bacteria, there is no need of oxygen. The granules and vesicles of phagocytes secrete various enzymes which are hydrolytic proteins. Those proteins are bacteriocidal in nature according to their modes of action

Oxygen dependent

There is requirement of oxygen for such kind of intracellular killing of bacteria. It is also termed as “Respiratory Burst” as requirement of glucose and oxygen get increased after phagocytosis. As a result, oxygen containing bacteriocidal radicals are produced.

49. Goodpasture's syndrome 2 marks

Goodpasture syndrome is a group of acute illnesses that affects the lungs and kidneys. It involves an autoimmune disorder. Normally, the immune system makes antibodies to fight off germs. But with Goodpasture syndrome, the immune system mistakenly makes antibodies that attack the lungs and kidneys. This condition can quickly progress to an inflammation of the kidneys (glomerulonephritis) and kidney failure. It can be fatal if not quickly diagnosed and treated.

50. Difference between BCR and TCR (3)

BCR

During the process of antibody formation, B-cells recognize the antigen with its specific receptor which is also called as B-cell receptor (BCR). After binding with specific antigen with BCR on B-cell, there is transformation of B-cells into a specialized variety of cells which are called as plasma cells. Plasma cells are also called as antibody secreting cells as these secrete antibodies.

TCR

T cell receptor (TCR) recognizes mainly those antigens which are protein in nature. Those antigens are processed in fragmented form before recognition by TCR. Unlike BCR, TCR cannot recognize soluble form of antigens. The processed proteins (antigens) are recognized by TCR after association with a complex called as major histocompatibility complex (MHC).

51. domains of immunoglobulins structure

Domains

These are folded regions which contain an intra-chain disulfide bond. Usually, heavy chains contain four domains while light chains composed on two domains

52. physical barriers (2)

The skin, mucous membranes, and endothelia throughout the body serve as physical barriers that prevent microbes from reaching potential sites of infection. Tight cell junctions in these tissues prevent microbes from passing through.

53. Role of Th1 and Th2 cells in immune response (10 marks)

• Th1 cells

These cells release cytokine like $\text{IFN}\gamma$ upon activation which then activate macrophages. In turn, macrophages participate in the generation & differentiation of CTL for cell mediated immunity. However, $\text{IFN}\gamma$ acts as inhibitory cytokine for Th2 (Fig.15.5)

• Th2 cells

Upon activation, Th2 cells secrete cytokines like IL-4 & IL-5 which further activate B-cells for humoral immunity. This kind of humoral response is also involved in the generation of IgE antibodies through the process of class switching. Moreover, Th2 cells are also involved in the activation of mast cells and eosinophils for providing immunity against allergen and parasitic infections. Th2 cell also secrete IL-10 which is inhibitory for $\text{IFN}\gamma$ production from Th1 cells

54. Process of cytotoxic T cells to kill viral infected cells (10 marks)

The cytotoxic T-Lymphocytes are also called as CTLs and $\text{CD8}^{\text{+ve}}$. These cells are not mature after exiting thymus where these have functional TCR which can recognize antigens but cannot

kill the target cells. There is need of differentiation for full activation of CTLs. CTL differentiates from Pre-CTLs in response to following two signals

1. The specific antigen in context of Class I MHC
2. The cytokines produced from T-helper 1 (Th1) cells especially IL-2 & INF- γ (Fig.15.1)

CTLs cause the killing of target cells. This killing is antigen specific where target cell must bear same antigen with class I MHC molecules.

55. Diapedesis

Vasodilators produced at the site of infection cause the junctions between endothelial cells to loosen and the phagocytes then cross the endothelial barrier by “squeezing” between the endothelial cells in a process called **diapedesis**.

56. Proinflammatory cytokines

Pro-Inflammatory cytokines

These are involved in up regulation of inflammatory process. These are predominantly produced from activated macrophages and Th cells like IL-1 β , IL-6 & TNF- α , IL-12 and INF- γ . These have role in mediating innate immune response. Furthermore, various inflammatory diseases like atherosclerosis, cancer and depression also occur due to these cytokines.

57. function of complement protein

Functions of Complement Proteins

Complement system provides specific as well as non-specific resistance against infections. This kind of resistance is exhibited with following two important functions

• Primary function

This function of complement proteins is considered primary as these activated proteins are used to kill or lyse infectious agents like bacteria. For this complement proteins get deposited on the surface of infectious agents which make hole in those cells for their lysis or killing.

• Secondary or Effector functions

These functions of complement proteins effect on other functions of immune system. These functions of complement proteins are as follows

- 1) Opsonization
- 2) Chemotaxis
- 3) Anaphylaxis

58. Name the component or barrier of immunity against infections (2)

Innate immune system is the first line of defense against invading pathogens. It is comprised of following three important components or barriers

- 1) Anatomical barriers
- 2) Humoral (Secretory) barriers
- 3) Cellular barriers

59. Define lymphokine activated killer cells (2)

Lymphokine activated killer (LAK) cells are NK cells which are differentiated by continuous exposure to IL-2 and interferon (Fig.15.10). Upon activation, LAK cells kill the transformed and malignant cells. This killing process is selective for sparing normal cells but needs cell-to-cell interaction.

60. How the diversity of T cell receptors is generated at genetic level (2)

TCR diversity is generated by the random and imprecise rearrangements of the V and J segments of the TCR alpha (TCRA) and V, D, and J segments of the TCR beta (TCRB) genes in the thymus. Thymic production of T cells is the sole mechanism to generate TCR diversity.

61. How cellular immunity produces protection (3)

Cellular mediated immunity provides protection by following ways

- **T-cell immunity**

In this form of protection, cytotoxic T-cells (CTL) are activated for the killing of infected and transformed cells.

- **By activating macrophages**

By this way, macrophages are activated for destroying various pathogens.

- **By activating natural killer (NK) cells**

In this way of protection, NK cells are activated for killing of transformed and virally infected cells.

62. Neutralization (3)

Neutralization

The pathogens or foreign substances are blocked from binding with their targets as antibodies bind with pathogens. For example, antibodies to bacterial toxins can prevent the binding of the toxin to host cells, thereby rendering the toxin ineffective. Similarly, antibody binding to a virus or bacterial pathogen can block the attachment of the pathogen to its target cell, thereby preventing infection or colonization.

63. What are the outcomes of inflammation (5)

Outcomes of Inflammation

In response of inflammatory process, following outcomes occur:

- ✓ There is restoration of normal tissue
- ✓ Large amount of tissue is destroyed which is called as fibrosis
- ✓ There is also pus formation which is also termed as abscess formation
- ✓ Inflammation can be prolonged for longer time points which is also called as chronic inflammation

64. Steps involving in phagocytosis (5)

Phagocytosis

After attachment of a bacterium, the phagocyte begins to extend pseudopods around the bacterium. The pseudopods eventually surround the bacterium and engulf it, and the bacterium is enclosed in a phagosome. During phagocytosis the granules or lysosomes of the phagocyte fuse with the phagosome and empty their contents. The result is a bacterium engulfed in a phagolysosome which contains the contents of the granules or lysosomes.

65. Define cytokines + mechanisms of action+ cytokine of innate immune system

Cytokines

These are secretory molecules which act as mediators between immune cells. Il-1 induces fever and the production of acute phase proteins, some of which are antimicrobial because they can opsonize bacteria

66. Activation of Cytotoxic T-cells

The surface of a cytotoxic T-cell has something called a CD8 receptor. The CD8 receptor interacts with cells called major histocompatibility complex (MHC) Class I molecules to recognize when healthy cells are infected.

When the CD8 receptor recognizes an infected cell, it activates the cytotoxic T-cells. The cytotoxic T-cells create molecules designed to destroy the infection.

67. Delayed type hypersensitivity (type IV) 5 marks

These reactions are also termed as cell mediated hypersensitivity. Due to more time requirement like 48-72 hours, these reactions are also called as delayed type of hypersensitivity.

Immunologically, T-cells, Macrophages and monocytes are involved in these reactions. In these reactions, the response is against intracellular pathogens that's why these reactions are antibody independent in nature. Lymphocytes like CD8 (Cytotoxic) T-cells cause direct damage to target cells. Likewise, CD4 (Helper) T-cells for example Th1 cells recognize foreign antigen with MHC class II. Moreover, the secretion of Cytokines, activation of macrophages and Inflammation increase the magnitude of tissue injury & damage

68. What is tolerance? Why microorganisms have high tolerance? 5 marks (REPEAT)

The specific immunological unresponsiveness or non-reactivity to an antigen resulting from a previous exposure to same antigen is called as immunological tolerance. This kind of unresponsiveness is mostly against self-antigens. This form of non-reactivity can be induced for non-self (foreign) antigen. The antigens which induce tolerance are called as tolerogen. Physiologically, there is no immune response against self-antigens which is also termed as self-tolerance. On the other hand, if immune response is generated against self-antigens is called as autoimmunity (Pathology).

Most bacteria and viruses develop tolerance which is the way to exploit or evading host immune system for example in Lepromatus type of leprosy there is tolerance against M. leprae as a result no immune response is generated. Tolerance to tissue and cell antigens can be induced artificially for example by injecting hemopoietic stem cells in neonatal or severely immunocompromised animals. Likewise, the generation of chimeras animals through transferring of allogeneic primary lymphoid tissues at early developmental life of animal. These are the ways of inducing tolerance against allogeneic tissues.

69. What is lymphocytic recirculation (3)

Naive (virgin) lymphocytes enter the lymph nodes from the blood via High Endothelial Venules (HEVs). Homing receptors on the lymphocytes direct the cells to the HEVs. It is estimated that about 1-2% lymphocytes recirculate every hour. In the lymph nodes, lymphocytes with the appropriate antigen receptor encounter antigen, which has been transported to the lymph nodes by dendritic cells or macrophages. This process of activation of lymphocytes is called as priming and lymphocytes are called primed lymphocytes. After activation the lymphocytes express new receptors that allow the cells to leave the lymph node and reenter the circulation. Receptors on the activated lymphocytes recognize cell adhesion molecules expressed on endothelial cells near the site of an infection and chemokines produced at the infection site help attract the activated cells.

70. Co stimulatory molecules of B cells in case of T dependent cells (3)

For signal 2, the co-stimulatory molecules are also required for activation of T-cells e.g C28 is a co-stimulatory molecule on T-cell surface & its ligand exists on APC which is known as B7-1. If there is lack of co-stimulation, then there is no activation of T-cells occur and this process is called as cellular anergy. For the full activation of T-cells, the engagement of Immunological synapse is mandatory. As a result the antigen presenting cells (APC) must possess & present peptides to Tcells.

71. SubTypes of Th cells (2)

There are two major subtypes of Th cells

- 1) Th1 cells

2) Th2 cells

• Th1 cells

These cells release cytokine like $\text{IFN}\gamma$ upon activation which then activate macrophages. In turn, macrophages participate in the generation & differentiation of CTL for cell mediated immunity. However, $\text{IFN}\gamma$ acts as inhibitory cytokine for Th2 (Fig.15.5)

• Th2 cells

Upon activation, Th2 cells secrete cytokines like IL-4 & IL-5 which further activate B-cells for humoral immunity. This kind of humoral response is also involved in the generation of IgE antibodies through the process of class switching. Moreover, Th2 cells are also involved in the activation of mast cells and eosinophils for providing immunity against allergen and parasitic infections. Th2 cell also secrete IL-10 which is inhibitory for $\text{IFN}\gamma$ production from Th1 cells

72. X linked hyper IgM immunodeficiency (2)

X-linked hyper IgM -Immunodeficiency

This form of B-cell associated immunodeficiency has very high levels of IgM. As a result, the levels of IgG & IgA get reduced. The underlying defect in this condition is in class switching due to defect in CD40 ligand on CD4 T-cells. The patients are susceptible to pyogenic infections like bacterial infections. This condition can be treated with intravenous gamma globulins injections.

73. Anaphylatoxins

Anaphylaxis is the form of immune response in which exaggerated response is generated against allergen. The molecules which produced such response are called as anaphylatoxins. The complement proteins like C4a, C3a & C5a act as potent anaphylatoxins. These molecules cause basophils and mast cells degranulation as a result various mediators or cytokines are generated. Those mediators involved in smooth muscles contraction, vasodilatation and bronchoconstriction.

74. Rejection of transplants

Clinically the significance of MHC is in tissue transplantation. The cells and tissues are transplanted for the treatment of various diseases. The immune response is generated against transplanted tissue which is called as rejection or destruction of transplant. Immune response can be of following two types based on the kind of tissue rejection

1) Host vs Graft Rejection

In this kind of transplant rejection, there are antigens on the surface of graft which are being recognized as foreign by host immune system

2) Graft vs Host Rejection

In this form of transplant rejection, the lymphoid (immune) tissues in graft recognize host immune system as foreign as a result immune response is mounted

75. Disease of endogenous antigens

The example of endogenous antigens induced cytotoxic hypersensitivity include erythroblastosis fetalis which is also called as hemolytic disease of newborns (HDN). This condition occurs due to Rh incompatibility between mother & fetus. As a result, there is the development of anti-Rh antibodies in maternal serum (IgG) which has the ability of crossing placenta in subsequent pregnancy.

76. Chemotactic agents for Leucocytes chemotaxis (2)

Leucocytes provide response against infectious agents by

- Chemotaxis

This is the movement of circulating phagocytes towards the site of infection by various chemicals also called as chemotactic agents for example save our soul (SOS) signals from bacteria like N-formylmethionine containing peptides. The clotting system peptides and complement proteins also serve as chemotactic agents. For macrophages, cytokines are considered as good chemotactic agents

77. Signs of inflammation (3)

Inflammatory response is exhibited in the form of undesirable effects which are in the form of following signs of inflammation.

1. Dolor (Pain)
2. Calor (Heat)
3. Rubor (Redness)
4. Tumor (Swelling)
5. Functio laesa (Loss of function)

78. If new virus/bacteria enter the body of an adult human, which type of immune response is generated? Explain briefly. (3)

The immune response is how your body recognizes and defends itself against bacteria, viruses, and substances that appear foreign and harmful.

The immune system protects the body from possibly harmful substances by recognizing and responding to antigens. Antigens are substances (usually proteins) on the surface of cells, viruses, fungi, or bacteria. Nonliving substances such as toxins, chemicals, drugs, and foreign particles (such as a splinter) can also be antigens. The immune system recognizes and destroys, or tries to destroy, substances that contain antigens.

79. Adverse effects of Inflammation (5)

Inflammation is associated with general flu like symptoms including

- Fever
- Chills
- Fatigue/Lethargy/Loss of energy
- Headaches
- Loss of appetite
- Body pains
- Muscle stiffness

80. What is Clonal selection? What are the principles for this clonal selection?

Clonal Selection of Lymphocytes

Lymphocytes either B or T carry the unique specificity with their receptor. The specificity determines the nature of lymphocytes. There is a vast array of T & B lymphocytes with their unique receptor for particular antigenic determinants. The selection of each repertoire of lymphocytes with unique specificity occurs by selection of each specific clone of B or T-lymphocytes with unique BCR & TCR respectively which is termed as clonal selection theory. The clonal selection hypothesis states that the germline encodes many different antigen receptors - one for each antigenic determinant to which an individual will be capable of mounting an immune response. Antigen selects those clones of cells that have the appropriate receptor. There are following four basic principles of the clonal selection theory

1. Each lymphocyte bears a single type of receptor with a unique specificity.
2. Interaction between a foreign molecule and a lymphocyte receptor capable of binding that molecule with a high affinity leads to lymphocyte activation.
3. The differentiated effector cells derived from an activated lymphocyte will bear receptors of an identical specificity to those of the parental cell from which that lymphocyte was derived.
4. Lymphocytes bearing receptors for self-molecules are deleted at an early stage in lymphoid cell development and are therefore absent from the repertoire of mature lymphocytes.

81. Defects of Deficiency of IgA?

IgA deficiency

There is deficiency of IgA in this disorder as a result individuals are susceptible to mucosal surface infections like GIT, eye & nasopharynx.

82. Mode of vaccination**Types of Vaccination**

There are two modes of providing specific protection against mostly common & damaging pathogens. These modes of immunization can be natural or artificial in their nature. Following are the types of immunization

- 1) Active Immunization
- 2) Passive Immunization

83. What is directly or indirect ELISA and application of ELISA (Long)

These are the kind of antigen & antibody reactions which can be detected on the basis of substrate utilization by enzyme. These reactions are also called as solid phase Immunoassay as these reactions occur on the solid surface e.g surface of plastic. Like other immune reactions, this is also used for determining either unknown antigen or antibody. Followings are the two types of ELISA based on whether antigen or antibody is determined

- 1) Direct ELISA
- 2) Indirect or Sandwich ELISA

Direct ELISA

In this type of ELISA, unknown antibody can be determined in patient's serum. Principally, known antigen is coated on solid phase which can be a Glass or plastic surface. Patient's serum containing unknown antibody is added for reacting with the specific coated antigen. Afterward, another antibody which is pre-coated with an enzyme e.g horseradish peroxidase is usually used. Then the activity of enzyme is determined by adding its substrate i.e hydrogen peroxide in the form of colour change in the reaction mixture. Moreover, the intensity of color determine the concentration of unknown antibody in the patient's serum indirectly.

Indirect ELISA

In this type of ELISA, unknown antigen can be determined in patient's serum. Principally, known antibody against the unknown antigen is coated on solid phase which can be a Glass or plastic surface. Patient's serum containing unknown antigen is added for reacting with the specific coated antibody. Afterward, another antibody which is pre-coated with an enzyme e.g horseradish peroxidase is usually used, thus the unknown antigen get sandwiched between two antibodies, that's why it is also called as sandwiched ELISA. Then the activity of enzyme is determined by adding its substrate i.e hydrogen peroxide in the form of color change in the reaction mixture. Moreover, the intensity of color determines the concentration of unknown antibody in the patient's serum indirectly.

Applications of ELISA

Followings are the significant applications of ELISA

- It is used for determining viral antigens in patient's serum e.g Hepatitis B surface Antigens (HbsAG)
- Also used for the determination of antibodies against viruses e.g Anti Hepatitis C Virus (HCV) antibodies
- Also used for estimating the levels of various serum cytokine e.g Interleukin-1 (IL-1), Tumor necrotic factor alpha (TNF-a) etc.

84. Define Allograft, Isograft ,and Xenograft (3)

Isograft

This is a form of tissue which is transplanted between individuals of same genetic makeup like identical twins.

Allograft

This form of transplantation which can be from one person to other non-identical person

Xenograft

This kind of transplantation is across the species for example from animal to human

85. How HLA help in transplantation(2)

HLA matching promotes the growth and development of new healthy blood cells (called engraftment) and reduces the risk of a post-transplant complication called graft-versus-host (GVHD) disease.

86. Mast cell degranulation

Mast cells have the capacity to release a multitude of pro-inflammatory mediators. 3. The immediate response upon mast cell activation to an appropriate stimulus is degranulation; a process characterized by the extrusion of cytoplasmic granule contents into the extracellular space by the process of exocytosis.

Past papers

1. Functions of Cytokines? 5 Marks

- Used as biological response modifier for treatment of various diseases
- Cytokine is a general term but more specific term used according to their cell of origin
- Monokines: produced by mononuclear phagocytic cells
- Lymphokines: produced by activated lymphocytes
- Cytokines function as cascade signaling in the form of cytokine network
- Cytokines are considered as redundant
- Redundancy: is due to the nature of cytokine receptors

2. Subregion of variable region of antibody? 5 Marks

Variable region

- Present on heavy & light chains of Immunoglobulins Two sub-regions

1) Hypervariable (HVR)

- Based on variability of amino acid sequences
- Complementarity determining region

2) Framework Regions

- Regions between hypervariable regions
- Groups & sub-groups of immunoglobulins
- Products of different variable region genes

3. Types of Autoimmunity? 2 Marks

Autoimmunity:

Classification based on tissues or organs involved two following categories

- Organ-specific autoimmunity
- Non-organ Specific autoimmunity

4. Effector Function of complement protein? 3 Marks

Effector Functions of Complement Proteins

- Opsonization (Opsonins)
- Chemoattraction (Chemoattractants)
- Anaphylaxis (Anaphylatoxins)
- Pro-Inflammatory

5. Primary and secondary antibody response? 2 Marks

Primary Ab response

- First kind of antibody response against antigen
- Major class of antibody is IgM
- Ab are with less affinity & less persistent

Secondary Ab response

- Second kind of antibody response against antigen
- Major class is IgG but may IgE or IgA but less IgM

6. Inflammation and its role in body? 3 Marks

Inflammation is the body's attempt: To protect the body from harmful stimulus. Complex Tissue response

- Part of body's immune system

- Inflammatory response: series of events for inflammation
- For wound healing
- For clearance of infections

7. What is Opsonization? 2 Marks

Opsonization: some complement proteins are good opsonins. C3b, iC3b and C4b attach to microorganism first. This complex of complement protein binds with complement receptor on the surface of phagocytic cell.

8. Write two main types of tumor associated antigens. 2 Marks

There are two main types of tumor associated antigens

- 1) Tumor specific transplantation antigens
- 2) Tumor Associated transplantation antigens

9. Differentiate between Organ Specific autoimmunity and Non-Organ Specific autoimmunity. 2 Marks

Organ Specific autoimmunity

- Immune response against organ associated specific antigens
- Antibodies against organ associated antigens damage the organ
- Target organs
- Skin
- Thyroid gland
- Muscles

Non-Organ Specific autoimmunity

- Immune response not against organ associated specific antigens
- Antibodies against not organ associated antigens
- Systemic autoimmune diseases
- Mainly target organs: skin, joints, soft tissues etc.

10. Write down the mechanism of Oxygen independent killing? 3 Marks

Oxygen Independent

- No need of oxygen for such kind of intracellular killing of bacteria
- Granules & Vesicles of phagocytes secrete hydrolytic proteins
- Those proteins are bacteriocidal in nature according to their modes of action

Effector Molecule Function:

Cationic proteins (including cathepsin): Damage to microbial membranes Lysozyme: Splits mucopolysaccharide in bacterial cell wall.

Lactoferrin: Deprives proliferating bacteria of iron

Proteolytic and hydrolytic enzymes: Digestion of killed organisms

11. How the fragment of antibodies is obtained? Briefly elucide the properties of the fragment? 5 Marks

Antibodies Fragments

- Produced by proteolytic enzymes i.e papain
- Two fragments are produced which have structure & function relationship

1) Fab

2) Fc

A. Fab

- Fragment for antigen binding (Fab)
- Papain causes the cleavage at hinge regions to produce two identical fragments containing one heavy & one light chain
- Having monovalent valency for antigen binding

B. Fc

- Crystalizable Fragment (Fc)
- Contain two heavy chains with CH2 & CH3 domains
- Performed effector functions

12. Applications of Agglutination reaction? 2 Marks

Applications of Agglutination

- Blood grouping: ABO & Rh for both either on the basis of antigens of RBC or antibodies of serum
- Assessing bacterial infections e.g Widal test for diagnosis of Typhoid fever

13. Tolerance is achieved by natural and artificial? 2 Marks

Naturally: In case of autoimmune disorders e.g Rheumatoid Arthritis, SLE etc.

Artificially: exposure to immunosuppressive drugs or X-ray irradiation e.g in case of experimental animals for bone marrow transplantation.

14. Usage of Monoclonal antibodies? 3 Marks

- Usage in prevention, diagnosis & treatment of disease
- Immunophenotyping of Immune cells: Monoclonal Ab used against cell surface molecules like Cluster of Differentiation (CD)
- Typing of various tumors e.g Leukemia • Immunohistochemistry of solid tumors
- Use in Immunotherapy: Against various tumor marker e.g CD20 B-Cell lymphoma
- Use in Fluorescence Activated cell sorting (FACS)

15. Cytokines in adaptive immune system? 5 Marks

Cytokines which act primarily for Adaptive (Specific) immune system

1) IL-2

- Produced by Th cells but Tc also produced in lesser extent
- Stimulate B-cell division Can also activate NK cells s& monocytes
- IL-2 acts in autocrine manner on T-cells
- Activation of T-cells result in expression of IL2R & production of IL-2
- IL-2 binds to IL-2R and promotes cell division
- IL-2R decay if T-cells are not activated & eventually proliferative phase ends

2) IL-4

- IL-4 is produced by macrophages & Th2 cells
- Promotes the development of Th2 from naïve Th cells
- Differentiated Th2 cells result in the production of antibodies
- IL-4 involves in the class switching of antibody into IgE.

3) IL-5

- IL-4 is produced by Th2 cells
- Promotes the development of B-cells & eosinophiles
- Also activates mature eosinophiles

4) TGF- β

- TGF- β is produced by T cells and many other cells types
- Acts as inhibitory cytokine
- Inhibits the proliferation of T-cells & activation of macrophages
- Also acts on neutrophils & endothelial cells to block the effects of proinflammatory cytokines

16. Clonal Selection of Lymphocyte. 5 Marks

- T-Cell Receptors (TCR)
- Receptor Specificity
- Lymphocyte Repertoire
- After maturation, selection of lymphocytes which are non-reactive to self-antigens

Four principles

- 1) Unique Receptor Specificity
- 2) Lymphocyte Activation: Interaction with foreign molecule
- 3) Differentiation of lymphocyte: from single clone
- 4) Selection of lymphocytes bearing receptors against non-self-molecules

17. Immunization. 2 Marks

Ways of providing specific protection against many common & damaging pathogens.
Stimulation of organism's or individual's immune system.

18. Modes of Immunization. 2 Marks

These modes of immunization can be natural or artificial in their nature

- 1) Active Immunization
- 1) Passive Immunization

19. T-cell Receptor and B-Cell Receptor? 3 Marks

T-cell Receptor:

Surface receptor for binding with Ag bounded with MHC. Like immunoglobulin: Part of Ig Superfamily. TCR is a heterodimer surface receptor. Composed of two chains of equal length

1. A Chain
2. B Chain

Short Cytoplasmic tail: Not sufficient for signal transduction. Transmembrane region: for anchoring cell membrane (due to hydrophobic amino acids).

B-Cell Receptor:

Cross linking of BCR by Ag. By cross linking, there is activation of Src for cellular signaling. Src phosphorylates tyrosine on ITAM of $I\alpha$ & β . Phosphorylated ITAM act as docking site for Syk. Downstream signaling for antibody synthesis.

20. T-cell antigen Kinetic Process? 3 Marks

T-independent Ag

- Mainly IgM production
- All those phases in primary response
- No secondary response as same phases in second time exposure to Ag
- No memory response

21. Drug induced hemolytic anemia? 5 Marks

- Also known as cytotoxic hypersensitivity
- Effect various organs & tissues (Tissue specific)
- Antigens are usually endogenous
- Exogenous: Chemicals in the form of haptens
- Binds to the surface of cell membrane
- Antibodies against antigens: IgM or IgG

Example of exogenous, antigen: Drug induced hemolytic anemia, granulocytopenia & thrombocytopenia

- Ag-Ab reaction on the surface of RBC
- Complement fixation on immune complex on RBC surface
- Reaction time is minutes to hour
- Lysis of RBC

22. What type of Immune Response is Elicited by alloantigens? 2 marks

Alloantigens elicit both cell-mediated and humoral immune responses.

23. Write 2 diseases for which inactive bacterial cell or viral particles are used for vaccines? 2marks

Inactivated bacterial cell

1-Anthrax 2-cholera 3-plague

Viral particle

1-hepatitis A 2-influenza 3-polio

24. Write Disease for which live attenuated vaccines are used. 3

Tuberculosis, typhoid, measles, polio, rotavirus, chickenpox, yellow fever.

25. Write 6 symptoms of GVHR ? 3

- ✓ Skin rash ✓ Emaciation (becoming thin) ✓ Retarded growth ✓ Diarrhoea ✓ Hepatomegaly
- ✓ Splenomegaly ✓ Increase in bilirubin production ✓ Bile ducts are damaged ✓ anaemia

26. How a capsular antigen serves as vaccine, which cell are activated by these antigens give example ? 5

Capsular polysaccharides

- The virulence of some pathogenic bacteria depends primarily on the anti phagocytic properties of their hydrophilic polysaccharide capsule.
- Coating of the capsule with antibodies and or complement greatly increases the ability of macrophages and neutrophils to phagocytose such pathogens.
- The current vaccine for *Streptococcus pneumoniae*, which causes pneumococcal pneumonia, consists of 23 antigenically different capsular polysaccharides.

27. What are cytokines and its types? 5

Cytokines are a diverse group of non-antibody proteins released by cells that act as intercellular mediators, especially in immune processes

- Low molecular weight soluble proteins (polypeptides) (Less than 30kD)

- produced in response to microbes and other antigens

Types

1. Monokines - produced by mononuclear phagocytes (monocytes)
2. Lymphokines - produced by activated T cells, primarily helper T cells
3. Interleukins - cytokines made by one leukocyte and acting on other leukocytes
4. Chemokines-cytokines with chemotactic activities

28. Explain Function, mode of action and properties of cytokines? 10

Function

Cytokines classified according to their biologic actions into three groups:

1) Mediators and regulators of innate immunity

- Produced by activated macrophages and NK cells in response to microbial infection
- They act mainly on endothelial cells and leukocytes to stimulate the early inflammatory response to microbes

2) Mediators and regulators of acquired immunity

- Produced mainly by T lymphocytes in response to specific recognition of foreign antigens
- They include IL-2, IL-4, IL-5,, IL-13, IFN, Transforming growth factor- β (TGF- β) and lymphotoxin (TNF- β)

3) Stimulators of Haematopoiesis

- Produced by bone marrow, leukocytes
- Stimulate growth and differentiation of leukocytes
- Stem cell factors, IL-3, IL-7, GM-CSF

Mode of action

- Autocrine
 - act on the same cell that has secreted cytokine
- Paracrine
 - act on a nearby cell
- Endocrine
 - act on a distant cell reached through the circulation

Properties

1. Produced by cells involved in both natural and specific immunity
2. Mediate and regulate immune and inflammatory responses

3. Secretion is brief and limited not stored as pre-formed molecules synthesis is initiated by new short-lived gene transcription
4. Pleiotropic -different cell types to secrete the same cytokine or for a single cytokine to act on several different cell types
5. Redundancy -similar functions can be stimulated by different cytokines. Receptors for cytokines are heterodimers (sometimes heterotrimers) that can be grouped into families in which one subunit is common to all members of a given family
6. Often influence the synthesis of other cytokines. They can produce cascades, or enhance or suppress production of other cytokines They exert positive or negative regulatory mechanisms for immune inflammatory responses
7. Often influence the action of other cytokines. antagonistic -cytokines causing opposing activities. Additive/synergistic -two or more cytokines acting together
8. Bind to specific receptors on target cells with high affinity.
9. Cellular responses to cytokines are generally slow (hours), require new mRNA and protein synthesis

29. What is ADCC?

Antibody-Dependent Cell-Mediated Cytotoxicity (ADCC)

- Cells Capable of Cytotoxicity Express Fc Receptors
- Antibody Binds Target Cell, Cytotoxic Cells Bind Fc Portion Of Ab
- Antibody Provides The Specificity
- Examples Of Cells Capable Of ADCC
- Macrophages, NK, Neutrophils, eosinophils

30. Mechanism of hypersensitivity type III?

Mechanism of Type III Hypersensitivity

- Antigens combines with antibody within circulation and form immune complex
- Wherever in the body they deposited
- They activate compliment system
- Polymorphonuclear cells are attracted to the site
- Result in inflammation and tissue injury

31. Antibody and antigen?

Antibody is known as immunoglobulin, is a large, y-shaped protein produced mainly by plasma cells that is used by the immune system to neutralize pathogen such as pathogenic bacteria. A toxin or the foreign substance which induces an immune response in the body especially the production of antibodies.

32. Cell mediated responses name?

Note: this answer is not conformed

- Primary Function Of Cell Mediated Response
- Eliminate Intracellular Pathogens
- Eliminate Tumor Cells
- Both Antigen Specific And Non-specific cells Are Involved
- Ag Specific: Both Cytotoxic and Helper T cells • Non-specific: Macrophages (M), Neutrophils, NK
- Both Specific And Non-specific responses Require Cytokines

33. Define proto oncogene.

A proto-oncogene is a normal gene that can become an oncogene, either after mutation or increased expression. Proto-oncogenes code for proteins that help to regulate cell growth and differentiation. Proto-oncogenes are often involved in signal transduction and execution of mitogenic signals, usually through their protein products. Upon activation, a protooncogene (or its product) becomes a tumor inducing agent, an oncogene.

34. Define hypersensitivity.

Hypersensitivity (Immunological reaction) refers to undesirable immune reactions produced by the normal immune system. Hypersensitivity reactions: When an immune response result in exaggerated OR in appropriate reactions harmful to the host, the term hypersensitivity OR allergy used. Hypersensitivity reactions: four types; based on the mechanisms involved and time taken for the reaction, a particular clinical condition (disease) may involve more than one type of reaction.

35. Mechanism of hypersensitivity?

Mechanism of Type III Hypersensitivity: Antigens combines with antibody within circulation and form immune complex

- Wherever in the body they deposited
- They activate compliment system
- Polymorphonuclear cells are attracted to the site
- Result in inflammation and tissue injury Of the mechanism of type III (immune-complex mediated hypersensitivity

36. What you know about hypersensitivity type?

Antigens combine with antibody within circulation and form immune complex. Wherever in the body they deposited. They activate compliment system. Polymorphonuclear cells are attracted to the site. Result in inflammation and tissue injury.

37. Innate and adaptive immunity.

Inborn or innate immunity: It is present at birth; this is our First Line Of Defense. If any invader penetrates the body's first line of defense mechanism is the innate immune system. Components of innate immunity are epithelial barriers, cellular mechanisms, humoral mechanisms. Role of

innate immunity in stimulating adaptive immune response Adaptive Immune System (AIS) AIS has two components.

Humoral immunity is mediated by soluble antibody proteins produced by B lymphocytes that neutralize and kill pathogens/toxins (20%-30%).

Cellular immunity is mediated by T lymphocytes which target cells infected by a pathogen (60%-70%)

38. Cytokines and its types.

Cytokines: Cytokines are a diverse group of non-antibody proteins released by cells that act as intercellular mediators, especially in immune processes

- Low molecular weight soluble proteins (polypeptides) (Less than 30kD)
- produced in response to microbes and other antigens

Functional Categories of Cytokines:

Cytokines classified according to their biologic actions into three groups:

1) Mediators and regulators of innate immunity

- Produced by activated macrophages and NK cells in response to microbial infection
- they act mainly on endothelial cells and leukocytes to stimulate the early inflammatory response to microbes

2) Mediators and regulators of acquired immunity

- Produced mainly by T lymphocytes in response to specific recognition of foreign antigens
- They include IL-2, IL-4, IL-5,, IL-13, IFN, Transforming growth factor- β (TGF- β) and lymphotoxin (TNF- β)

3) Stimulators of haematopoiesis

- Produced by bone marrow, leukocytes
- Stimulate growth and differentiation of leukocytes
- Stem cell factors, IL-3, IL-7, GM-CSF

39. Signal of cell mediated receptor?

Cell signaling is part of any communication process that governs basic activities of cells and coordinates all cell actions. The ability of cells to perceive and correctly respond to their microenvironment is the basis of development, tissue repair, and immunity, as well as normal tissue homeostasis. Proteins responsible for detecting stimuli are generally termed receptors, although in some cases the term sensor is used. The changes elicited by ligand binding (or signal sensing) in a receptor give rise to a signaling cascade, which is a chain of biochemical events

along a signaling pathway. When signaling pathways interact with one another they form networks, which allow cellular responses to be coordinated, often by combinatorial signaling events.²¹ At the molecular level, such responses include changes in the transcription or translation of genes, and post-translational and conformational changes in proteins, as well as changes in their location.

40. Three names of vaccines which are live attenuated?

Tuberculosis Typhoid Rotavirus Varicella zoster (chickenpox) Yellow fever

41. 4. Explain the structure of B cell and T cell under microscope?

B Cells

BCRs bind intact antigens (like diphtheria toxoid, the protein introduced into your body in the DTP vaccine). These may be soluble molecules present in the extracellular fluid; or intact molecules that the B cell plucks from the surface of antigen presenting cells like macrophages and dendritic cells.

The bound antigen molecules are engulfed into the B cell by receptor-mediated endocytosis. The antigen is digested into fragments which are then displayed at the cell surface nestled inside a class II histocompatibility molecule.

Helper T cells specific for this structure (i.e., with complementary TCRs) bind the B cell and secrete lymphokines that: o stimulate the B cell to enter the cell cycle and develop, by repeated mitosis, into a clone of cells with identical BCRs; switch from synthesizing their BCRs as integral membrane proteins to a soluble version; differentiate into plasma cells that secrete these soluble BCRs, which we now call antibodies. T Cells The surface of each T cell also displays thousands of identical T cell receptors (TCRs). There are two types of T cells that differ in their TCR:

alpha/beta ($\alpha\beta$) T cells. Their TCR is a heterodimer of an alpha chain with a beta chain. Each chain has a variable (V) region and a constant (C) region. The V regions each contain 3 hypervariable regions that make up the antigen-binding site. gamma/delta T cells. Their TCR is also a heterodimer of a gamma chain paired with a delta chain.

The discussion that follows now concerns alpha/beta T cells. Gamma/delta T cells, which are less well understood, are discussed at the end.

The TCR (of alpha/beta T cells) binds a bimolecular complex displayed at the surface of some other cell called an antigen-presenting cell (APC). This complex consists of: a fragment of an antigen lying within the groove of a histocompatibility molecule

42. Mechanism of GVHR?

✓ Graft versus host reaction In some instance the graft tissue elicits an immune response against host antigen and that immune response is called graft versus host reaction

- ✓ Graft versus host reaction brings damage to host cells and host
- ✓ When grafted tissue has mature T cells, they will attack host tissue leading to GVHR.

Mechanism of GVHR:

- ✓ Graft lymphocytes aggregate in the host lymphoid organs
- ✓ Graft lymphocytes are stimulated by the host lymphocyte
- ✓ Stimulated lymphocytes of graft produce lymphokines
- ✓ Lymphokines activate host T-cell which produce polyclonal b-cell activation
- ✓ Activated b-cell react with the self-antigens and cause damage to the host cell

43. Define antibodies? and five antibody classis in mammals?

A blood protein produced in response to and counteracting a specific antigen. Antibodies combine chemically with substances which the body recognizes as alien, such as bacteria, viruses, and foreign substances in the blood.

IgA (immunoglobulin A): Human immunoglobulin A (IgA) is the second most common human immunoglobulin in serum. It is secreted in milk and is also the most prevalent Ig in secretions (e.g. tears, saliva and mucous). IgA is resistant to digestion and can activate the complement pathway when aggregated. It should also be noted both subclasses of IgA (IgA1 and IgA2) bind fragment crystallization (FC) receptor

IgD (immunoglobulin D): Expressed on the surface of mature B cells, human immunoglobulin D (IgD) works with IgM in B cell development. IgD is found in very low levels in serum and does not activate the complement pathway

IgE (immunoglobulin E): Human immunoglobulin E (IgE) is expressed on the surface of mature B cells. Human immunoglobulin IgE is the least abundant Ig in the serum and does not activate the complement pathway. Fragment crystallization (FC) receptors for IgE are found on eosinophils and IgE binds Fc receptors on mast cells and basophils even before interacting with antigen. As a result of its binding to basophils and mast cells, IgE is involved in allergic reactions. This happens when allergen is bound to IgE on cells and releases various pharmacological mediators which cause allergies.

IgG (immunoglobulin G): Human IgG (immunoglobulin G) is expressed on the surface of mature B cells and is the most prevalent Ig in serum and the major Ig in extravascular spaces. IgG1, IgG2 and IgG3 are complement activators, with IgG3 being the strongest. Human immunoglobulin subclasses IgG1 and IgG3 tend to strongly bind fragment crystallization (FC) receptors, whereas subclasses IgG2 and IgG4 bind weakly. IgG is also the only human immunoglobulin to pass from mother to fetus to transfer immunity.

IgM (immunoglobulin M): Human IgM (immunoglobulin M) is expressed on the surface of immature and mature B cells as monomers. IgM is the third most abundant human

immunoglobulin. IgM is also the first human immunoglobulin to be made by a fetus and virgin B cells which are challenged with antigen. IgM is a strong complement activator and agglutinator due to its pentameric structure and binds fragment crystallization (FC) receptors.

44. Memory cell and Nk cell?

Natural killer cells (also known as NK cells, K cells, and killer cells) are a type of lymphocyte (a white blood cell) and a component of innate immune system. NK cells play a major role in the host-rejection of both tumours and virally infected cells.

Memory cells: a long-lived lymphocyte capable of responding to a particular antigen on its reintroduction, long after the exposure that prompted its production

45. What is antigen?

In immunology, an antigen is a molecule capable of inducing an immune response (to produce an antibody) in the host organism. Sometimes antigens are part of the host itself in an autoimmune disease. Antigens are "targeted" by antibodies.

46. Name two diseases that are caused by bacterial or viral infection used as vaccine?

Otitis media is the official name for an infection or inflammation of the middle ear. Respiratory tract infections include sore throat, bronchitis, sinusitis, and pneumonia. Bacteria or viruses may be responsible for respiratory tract infections. Tuberculosis is a type of bacterial lower respiratory tract infection.

47. Name types of receptors of cytokines?

Type I receptor Type II Type III Type IV

48. Write the function of spleen?

The spleen plays multiple supporting roles in the body. It acts as a filter for blood as part of the immune system. Old red blood cells are recycled in the spleen, and platelets and white blood cells are stored there. The spleen also helps fight certain kinds of bacteria that cause pneumonia and meningitis.

49. Clinical symptoms of graft versus host reaction?

Clinical manifestations of acute GVHD include a classic maculopapular rash; persistent nausea and/or emesis; abdominal cramps with diarrhea; and a rising serum bilirubin concentration

50. What is allergic contact dermatitis?

Allergic contact dermatitis:

- Cell-mediated immune response
- Results in an intensely irritating skin rash
- Triggered by chemically modified skin proteins that the body regards as foreign

- A cellular, fluid-filled blisters develop in severe cases
- Can be treated with glucocorticoids

51. Write the function of lymph nodes?

Lymph nodes are major sites of B and T lymphocytes, and other white blood cells. Lymph nodes are important for the proper functioning of the immune system, acting as filters for foreign particles and cancer cells. Lymph nodes do not have a detoxification function, which is primarily dealt with by the liver and kidneys.

52. Signal of cell mediated receptor?

Receptor-mediated endocytosis (RME), also called clathrin-mediated endocytosis, is a process by which cells absorb metabolites, hormones, other proteins - and in some cases viruses - by the inward budding of plasma membrane vesicles containing proteins with receptor sites specific to the molecules being absorbed (endocytosis).

53. Cytokines and its types 5 mark:

- Cytokines are a diverse group of non-antibody proteins released by cells that act as intercellular mediators, especially in immune processes
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- produced in response to microbes and other antigens

54. Cytotoxic t cells? 3 marks

A cytotoxic T cell (also known as To, cytotoxic T lymphocyte, CTL, T-killer cell, cytolytic T cell, CD8+T-cell or killer T cell) is a T lymphocyte (a type of white blood cell) that kills cancer cells, cells that are infected (particularly with viruses), or cells that are damaged in other ways.

Cytokines function, mode of action and receptors types 10 mark

Cytokines are a diverse group of non-antibody proteins released by cells that act as intercellular mediators, especially in immune processes • Low molecular weight soluble proteins (polypeptides) (Less than 30KD) produced in response to microbes and other antigens

55. Cytokine types?

Cytokines:

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3) Stimulators of haematopoiesis

- Produced by bone marrow, leukocytes
- Stimulate growth and differentiation of leukocytes
- Stem cell factors, IL-3, IL-7, GM-CSF

56. What are t cells and two classes of t cells 2 mark

A T cell, or T lymphocyte, is a type of lymphocyte (a subtype of white blood cell) that plays a central role in cell-mediated immunity. T cells can be distinguished from other lymphocytes, such as B cells and natural killer cells, by the presence of a T-cell receptor on the cell surface. Types:

- T helper cells (TH cells) assist other white blood cells in immunologic processes, including maturation of B cells into plasma cells and memory B cells, and activation of cytotoxic T cells and macrophages.
- Regulatory (suppressor)
- Natural killer T cell.
- Mucosal associated invariant.
- Gamma delta T cells.

57. Hypersensitive reaction type 1 5 mark

- Commonly called allergy
- Mediated by IgE antibodies produced by plasma cells in response to stimulation of Th2 cells by an antigens.
- The antigens that stimulate it are called allergens (i.e. House dust, Pollens, Cosmetics, Insects, Clothing and Drug)
- Exposure may be ingested, inhalation, injection or direct contact.
- Type I hypersensitivity reactions can be systemic (e.g., systemic anaphylaxis) or localized to a specific target tissue or organ (e.g., allergic rhinitis, asthma).

58. What are interferons (2)

Interferons (IFNs) are a group of signaling proteins made and released by host cells in response to the presence of several pathogens, such as viruses, bacteria, parasites, and also tumor cells.

Interferons are named for their ability to "interfere with viral replication by protecting cells from virus infections.

59. What are effector helper T cells (3)

Effector cells are the relatively short-lived activated cells that defend the body in an immune response. Effector B cells are called plasma cells and secrete antibodies, and activated T cells include cytotoxic T cells and helper T cells, which carry out cell-mediated responses. Antibodies have several modes of action:

- i) they can block ligand-receptor interactions;
- ii) cause cell lysis through activation of complement-dependent cytotoxicity (CDC);
- iii) interact with Fc receptors on effector cells to engage dependent cellular cytotoxicity;
- iv) signal for ingestion of a pathogen

60. Write names and functions of MHC proteins (3)

MHC:

2 main structurally & functionally distinct classes of MHC proteins: class I MHC proteins, present foreign peptides to cytotoxic T cells, & class II present foreign peptides to helper T cells. The major histocompatibility complex (MHC) of genes consists of a linked set of genetic loci encoding many of the proteins involved in antigen presentation to T cells, most notably the MHC class I and class II glycoproteins (the MHC molecules) that present peptides to the T-cell receptor.

The importance of MHC proteins is that they allow T cells to distinguish self from non-self. In every cell in your body, antigens are constantly broken up and presented to passing T cells. Without this presentation, other aspects of the immune response cannot occur

61. Write functions of spleen (3)

Spleen: The spleen plays multiple supporting roles in the body. It acts as a filter for blood as part of the immune system. Old red blood cells are recycled in the spleen, and platelets and white blood cells are stored there. The spleen also helps fight certain kinds of bacteria that cause pneumonia and meningitis

62. How B cells form IgM antibody (5) Write the steps

Class Switching: B-Cell initially produces IgM (antibody's heavy chain is encoded by the first constant region gene, the μ segment) molecules, which recognise specific antigenic determinant. When B cell becomes a plasma cell another deletion occurs in the cell's DNA, which positions heavy-chain variable region gene next to a different constant region (γ , μ , α , or δ). The First Step Interaction of Helper T Cells with Antigen on A Cells, and the Production of Lymphokines. In this first step, growth and maturation factors. A clone of helper T cells can produce its own T Lymphokines. Produced by A Cells and the T-Cell independent Activation of A Cells to a factor. Production The set of γ

63. When mast cells are activated and leukocytes enter the site of injury (5)

Mast cells initiate inflammation by releasing biochemical mediators from preformed cytoplasmic granules and synthesizing other mediators in response to a stimulus activation: Physical injury, chemical agents, immunologic processes, and toll-like receptors

Chemical release in two ways: Degranulation and synthesis of lipid-derived chemical mediators
Histamine: released from mast cells causes dilation of capillaries and retraction of endothelial cells lining the capillaries, which increases vascular permeability deficiencies.

Impaired contraction: Contracture three layers of human defense barriers, innate immunity (inflammatory response), and adaptive immunity. Physical and mechanical barriers first barriers of defense prevent damage to the individual and prevent invasion by pathogens skin & mucous membranes biochemical barriers Antibacterial peptides in mucous secretions, perspiration, saliva, tears, and other secretions.

The Inflammatory Process: Inflammation is a rapid and nonspecific protective response to cellular injury from any cause. It occurs only in vascular tissue macroscopic hallmarks of inflammation redness, swelling, heat, pain, and loss of function of the inflamed tissues.

64. What are antigens?

While antigens are the "target" of antibodies, immunoglobulin-binding proteins "attack" antibodies. T-independent antigen - Polysaccharides (usually) that stimulate B cells directly. Immuno dominant antigens – Antigens that dominate (over all others from a pathogen) in their ability to produce an immune response.

65. Define humoral and cellular immunity?

Humoral immunity or humoral immunity is the aspect of immunity that is mediated by macromolecules found in extracellular fluids such as secreted antibodies, complement proteins, and certain antimicrobial peptides. Its aspects involving antibodies are often called antibody-mediated immunity.

Cell-mediated immunity is an immune response that does not involve antibodies, but rather involves the activation of phagocytes, antigen-specific cytotoxic T-lymphocytes, and the release of various cytokines in response to an antigen.

66. What is the activation of cytotoxic T cells?

Helper T cells are arguably the most important cells in adaptive immunity, as they are required for almost all adaptive immune responses. They not only help activate B cells to secrete antibodies and macrophages to destroy ingested microbes, but they also help activate cytotoxic T cells to kill infected target cells B cell and t cell structure: Lymphocytes are one of the five kinds of white blood cells or leukocytes), circulating in the blood. Although mature lymphocytes all look pretty much alike, they are extraordinarily diverse

The most abundant lymphocytes are

- B lymphocytes (often simply called B cells) and T lymphocytes (likewise called T cells).

B cells are produced in the bone marrow. The precursors of T cells are also produced in the bone marrow but leave the bone marrow and mature in the thymus. Each B cell and T cell is specific for a particular antigen. What this means is that each is able to bind to a particular molecular structure.

The specificity of binding resides in a receptor for antigen:

- The B cell receptor (BCR) for antigen and the T cell receptor (TCR) respectively. They have a unique binding site. This site binds to a portion of the antigen called an antigenic determinant or epitope. The binding, like that between an enzyme and its substrate, depends on complementarity of the surface of the receptor and the surface of the epitope. The binding occurs by non-covalent forces (again, like an enzyme binding to its substrate). The successful binding of the antigen receptor to the epitope if accompanied by additional signals, results in the stimulation of the cell to leave go and enter the cell cycle.

Repeated mitosis leads to the development of a clone of cells bearing the same antigen receptor;

67. AIS activation?

Adaptive Immune System (AIS)

AIS has two components. Humoral immunity is mediated by soluble antibody proteins produced by B lymphocytes that neutralize and kill pathogens/toxins (20%-30%). Cellular immunity is mediated by T lymphocytes which target cells infected by a pathogen (60%-70%). IS Activation IIS Phagocytic cells have receptors that have affinity for certain pathogen-associated molecules IIS programmed to be activated by specific pathogen molecules (Toll Receptors) IIS activates AIS a

Adaptive Immune Response AIR is highly specific to the pathogen that induced it. There are 2×10^{12} lymphocytes in the human body, making the immune system comparable in cell mass to the liver or brain. Cells of immune system:

- B Cells.
- Dendritic Cells
- Granulocytes.
- Innate Lymphoid Cells (ILCs)
- Megakaryocytes.
- Monocytes/Macrophages.
- Myeloid-derived Suppressor Cells (MDSC)
- Natural Killer (NK) Cells.

68. Lymph node function?

Lymph nodes are major sites of B and T lymphocytes, and other white blood cells. Lymph nodes are important for the proper functioning of the immune system, acting as filters for foreign particles and cancer

69. What is the function of Fc antibody in IgM 2

Placenta cells that are in contact with mothers blood have Fc receptors that are use to grab IgG and pass it to the fetus. MHC Genes are represented for? Which cells are included in it and which proteins encode the MHC genes are located within the human major histocompatibility complex (MHC) and code for different members of the class I and the class II human leukocyte antigen (HLA) families. ... The open and shaded boxes represent pseudogenes and coding genes respectively.

70. Functions of Thymus 3

The thymus gland, despite containing glandular tissue and producing several hormones, is much more closely associated with the immune system than with the endocrine system. The thymus serves a vital role in the training and development of T lymphocytes or T cells, an extremely important type of white blood cell. The thymus gland is located in the neck above the heart. The function of the thymus gland is to generate mature T lymphocytes (white blood cells that help the immune system fight off illness).

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